

Proposal of a modeling approach and a set of KPI to the drug supply chain within the hospital

Houda Mezouar, Abdellatif El afia, Radouane Chiheb, Fatima Ouzayd

Mohammed V University in Rabat

ENSIAS

Rabat, Morocco

houda_mezouar@um5.ac.ma, a.elafia@um5s.net.ma, craddouane@yahoo.fr, ouzayd@gmail.com

Abstract—The hospital is a complex system where many actors are involved; The challenges that this environment is facing justify the focus on this research subject. The contribution of the paper includes a proposition of a four-level model, that describes the hospital drug supply chain, based on the combination of the Supply Chain Operation Reference (SCOR) and Business Process Modelling Notation (BPMN), and the identification of a set of key performance indicators (KPI) that assess how well the drugs management process within the hospital are effective and efficient and how to improve them.

Keywords—Hospital drug supply chain; supply chain modeling; performance measurement; SCOR; BPMN;

I. INTRODUCTION

The hospital settings have the same characteristics as industrial environments with much greater complexity and a very uncertain process of care. The entrance of a patient in the hospital triggers a therapeutic management process in which the drug is an essential component. The drug circuit in health facility covers two circuits, the first circuit, clinic, is that of the patient's drug therapy hospitalized since its entry, when its treatment is considered, until discharged when a prescription is made. Within the hospital, the clinic circuit includes prescription stages, dispensing and administration. The second circuit, logistics, concerns the drug as a product, from purchase to its delivery in the care unit, and it joins the clinic circuit in the step of administering the drug to the patient. Because it is based on human factors, the drug circuit involves significant risks of errors. These risks are increased in a hospital environment that is subject to a performance obligation. In this context, planning, the tasks care scheduling, requires good modeling and a system of performance measurement to fill the hazards and risks that may arise and disrupt the smooth functioning of the hospital supply chain and to provide a decision support tool. Modeling and measurement of this system helps to understand how it works and that it is flattening may have several purposes: better understanding, better communication, performance evaluation, analysis of malfunctions of system behavior.

There are a variety of supply chain models, which address both the upstream and downstream sides, in this paper and in order to model the hospital drug supply chain we use Supply Chain Operation Reference (SCOR) as a reference model and

Business Process Modelling Notation (BPMN) as a modelling tool. The choice of SCOR as a reference model is motivated by the fact that it allows both the modeling and evaluation of the performance. BPMN is used to model the fourth level of SCOR since the reference does not allow it.

The rest of this paper is organized as follows: the next section briefly traces both the hospital logistics and the drug logistics. By doing so, we define the perimeter of this study; we focus on these logistics definitions and the various flows that mark them. Section 3 outlines existing works in research, which are related to the hospital supply chain, and those related to the modeling techniques. In Section 4, we define the Supply Chain Operations Reference (SCOR) and the Business Process Management (BPM). The modeling methodology and the different models are provided in Section 5. A set of key performance is presented in section 6. This paper is concluded in Section 7.

II. THE LITERATURE REVIEW

A. The hospital supply chain

Several authors, and from different point of views, have faced the healthcare field in a general way, and especially the issues connected with the hospital supply chain. We see through literature, that there are two aspects to the definition of hospital logistics. One aspect is based on the logistics activity and a second is based on the logistics flow. By focusing on the overlap of the different activities of hospital logistics, [1] identified two approaches for the hospital supply chain analysis. One related to internal activities such as inventory control, receiving and internal distribution. The other one concern relations with external partners (supplier and distributors) which focuses more on the procurement process, contract negotiations, supplier selection and storage strategies. Reference [2] defined the hospital supply chain by a complex system of processes, operations, activities and organizations involved in the discovery, development and manufacturing of medications and drugs. According to reference [3], the drug circuit in a health facility covers prescription, analysis and validation of the prescribing, dispensing, delivery, distribution and administration of the drug, pharmacy orders. There is the circuit called "physical", which includes such delivery, receiving, repackaging, preparation and transportation of the

drug, and a second circuit called "information" on medication management, which affects more the order and transmission of information on the drug between members of the healthcare team. Reference [4], discusses the nature of healthcare supply chains particularly focusing on dental implant. The authors have characterized the dental supply chain and the basic process flow of the supply chain specifically for dental implants, they show the lacking areas which actually differentiate between the high technology supply chain and a conventional supply chain. Securing the medicine circuit in the hospital is a public health priority that involves a collective approach. In this context, [5] operates an inventory of the different possible approaches to risk reduction in hospital, illustrated by the experience of Grenoble University Hospital in this area. Within this framework, the authors confirm that computerization of the medication circuit represents a major advance in terms of organization and security. Similarly, the contribution of new technologies enables the automation of the overall distribution at the central pharmacy and deployment of drug distribution automata care units. In the same context, care quality and safety, the authors of [6] propose to come to the main approaches to business process improvements that have been used in hospitals focusing specifically on one of the newest and most used today: Lean. Reference [7], presents a radio-frequency identification (RFID)-enabled traceability system for the management of consignment and high value products requiring item level traceability in a hospital environment. According to the authors, when the health care supply chain members are included in the replenishment process with EDI, many other benefits can be derived. Various other research works were interested in a holistic hospital logistics chain, which links the hospital to its suppliers, [8] for their part investigated the influence of hospital-supplier integration on its supply chain performance and the influence of relational assets (interorganizational IT integration, trust, and knowledge exchange) on hospital-supplier integration. The results show that a hospital that has a high level of logistical integration with its suppliers will yield greater performance within its supply chain. Reference [9], described the decision-making process in French public hospitals, using qualitative interviews that were conducted with 14 pharmacists in 14 hospitals in various regions of France. The study showed that few structures used detailed guidelines on request for adding a new drug, preparation of the evaluation report and decision-making, to establish transparency in drug selection. The authors concluded that the complexity of the process of drug selection imposes the development in all the structures of a formal approach for the management of such a process. In order to understand the current operations of healthcare industries, the authors of [10] developed the procedure for determining the optimal solutions of inventory lot-size, lead-time and total number of deliveries with the objective of achieving target customer service level of the hospital at minimum pharmaceutical company-hospital supply chain total cost.

B. The modeling techniques

Discrete event system modeling, including supply chain is for the scientific community a primary concern. Reference [11] finds that the modeling is an activity of the brain, and defines

the meta-models, frameworks and methods in this spirit. According to [12], the lack of integration of supply chain systems with enterprise systems has been highlighted as a cause of disruptions of the supply chains by several studies that is why the authors find that the modeling is important when designing supply chain systems. Many supply chain models exist, the authors of [13] present a systematic review of the quantitative and analytical models for managing supply chain risks, they used bibliometric and network analysis tools to analyze quantitative Supply Chain Risk Management literature, examine the evolution of this research area, and identify emerging trends. The authors of [14] confirm the large number of the supply chain modeling methods, according to them there are two modeling paradigms, centralized and distributed. The centralized approach is any process where decision-making is made by a central authority. The established method for the centralized approach is mathematical programming. Agent-based modeling (ABM) is considered the most typical distributed approach for analyzing the supply chain performance. In other researches, the modeling approaches are divided into modeling type (mathematical, simulation, and multi-agent) and modeling settings (linear, integer, dynamic and stochastic problem settings) [15]. One of the most important roles of supply chain modeling is the supply chain simulation which is a scientific method by which users employ a model to observe the operation of an entire supply chain and conduct "what-if" analysis for multiple scenarios. There are several kinds of supply chain simulation methods, including process-oriented, object-oriented, system dynamics-based, petri-net-based, High Level Architecture (HLA)-based, and agent-based simulation [16]. For [17], the main role of the modeling techniques is to make the supply chain sufficiently understandable to ensure product quality; the authors even consider that the modeling methods have an important role to play in pharmaceutical process development, by supplementing experimental studies in evaluation process sensitivity and operability. Reference [18], highlighted the ability of the modeling methods to describe the real world phenomena, the authors divided the supply chain models into four types: stochastic model, deterministic model, hybrid model, IT-driven model.

C. The Supply Chain Operations Reference Model (SCOR)

SCOR provides a common framework, standard terminology, common metrics, and best practices [19]. The reorganization of the supply chain with SCOR is chronologically done in three steps, firstly we capture the state "as-is" by modeling the current situation at all levels, and then we develop a "to-be" desired state. Secondly, a Benchmarking step positions the company in its competitive environment by quantifying the performance of its competitors and by establishing internal targets based on "best-in-class" results. At this stage, the SCOR model provides us a huge number of performance indicators. The last step for modeling by SCOR is the definition of the target model of the business by analyzing the best practices [21]. According to reference [20], the SCOR model consists of processes in three hierarchical levels. The first level consists of the five different process types: Source, Make, Deliver, Return, and Plan. At the second level, the model distinguishes between make-to-stock (MTS) products,

make-to-order (MTO) products, and engineer-to-order (ETO) products. The Make process (M) e.g. is divided into Make-to-stock (M1), Make-to-order (M2), and Engineer-to-order (M3). Source and Deliver follows the same terminology, with an extra process; Deliver retail products (D4). The Plan process contains the overall process Plan supply chain (P1) and one planning process for each of the first level processes. Source Return and Deliver Return are divided into three sub-processes; return of defective products, return of MRO (maintenance, repair, and overhaul) products, and return of excess products. The complete SCOR model also includes the enabling processes. The third level describes the underlying processes of the second one, and they are generic for all companies. So it is recommended to add a fourth level to represent tasks that describe the third level activities and which will be unique to each business [21].

D. The Business Process Management (BPM)

BPM is a process centric approach for improving performance that combines information technologies with process and governance methodologies. The core of the BPM project deployment is the process modeling that has to consider and define the following elements: process scope; process start; process activities and their interconnections; process numbers (measurable figures of its activities. For example, the resources needed and their allocation, as duration, number of staff, maximum load etc.), process key performance indicators (KPI), process end and process connections with other processes [22]. At the heart of BPM is the explicit representation of processes through the BPMN (Business Process Model and Notation), an Object Management Group (OMG) standard. This notation was specially developed to facilitate human interpretation by both technical and non-technical personnel, and improve process execution. Once the processes are defined, they can be simulated, executed, monitored, and analyzed. In fact BPM tools provide an integrated environment where users can design and run BPMN models [23]. The benefits of BPM can be split into five categories [10]: First, the understanding of the business transaction and the rules that govern the activity is improved. Secondly, operational efficiency is improved and costs are reduced. Third, consistency of operation and better quality are obtained. Fourth, operational flexibility for the rapid and continuous change will provide optimization. Finally, performance and reporting are improved.

III. THE PROPOSED MODELING APPROACH

The objectives that can be assigned to a model are firstly its explanation capacity, description or understanding (e.g. model for interpreting observed behaviors or share a universe of discourse) and other part, his ability to guide action (design, simulation, monitoring / control, etc.) [24]. . The strength of a modeling methodology lies in its ability to combine approaches, tools or methods of different origins and organizing their arrangement in a very specific purpose [25]. As part of our problem to describe and to make clear the various drug management processes within a hospital, the suggested modeling methodology for the hospital drug supply chain, is based on the combination of SCOR and BPMN (see Fig. 1).

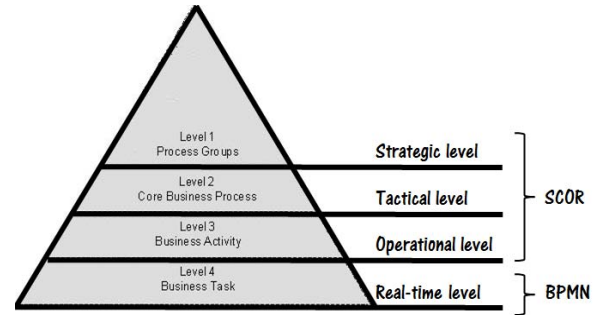


Fig. 1. The modeling methodology

The different levels, are organized in a descending order (that is, from the overall system, to decompose the latter into finer granularity subsystems). Within this methodology, the strategic level describe the scope and high level configuration of the hospital drug supply chain, the tactical level differentiate the strategies of the first level processes, the operational level describe the steps performed to execute the level 2 processes. The sequence in which these processes are executed influences the performance of the level two processes and the overall supply chain. The last level describe activities required to perform level three processes, it describe the detailed implementation of a process. We use SCOR for the modeling of the three first levels; the last one is done with the BPMN.

IV. THE MODELS

A. Strategic level

At this level, the company makes the basic strategic decisions, in our case we analyze the drugs supply chain within the hospital. As shown in Fig. 2 the supplier has two processes Deliver (D) describes the drugs delivery from the supplier to the hospital, and Return (R) describes the return permission of drugs from the hospital to the supplier. The hospital pharmacy has three processes, Source (S) describes drug procurement, Deliver (D) describes the drugs delivery from the hospital pharmacy to the patient, and Return (R) describes the drug return from the hospital to the supplier. The patient has the process Source (S) that describes in our case, receiving drugs by the patient. The process Plan (P) is the one that balance aggregate demand and supply to develop a course of action, which best meets sourcing, production, and delivery requirements.

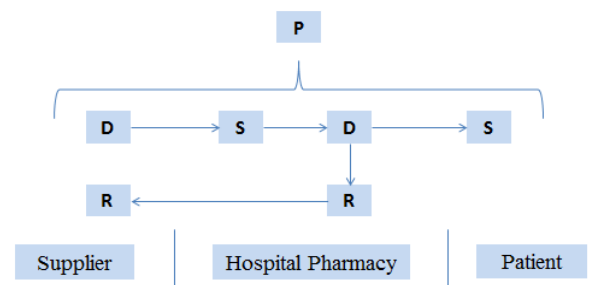


Fig. 2. The strategic level model

B. Tactical level

This level splits domains according to the configuration of the supply flow, and the results are Source Stocked Product (S1), and Deliver Stocked Products (D1). In our case, the return is concerning any drug of which the end user cannot benefit, the processes describe this return in SCOR are Source Return Defective Product (SR1) and Deliver Return Defective Product (DR1). We note that the Plan processes are Plan Supply Chain (P1), Plan Source (P2), Plan Deliver (P4), and Plan Return (P5). The modeling of this level is shown in Fig. 3.

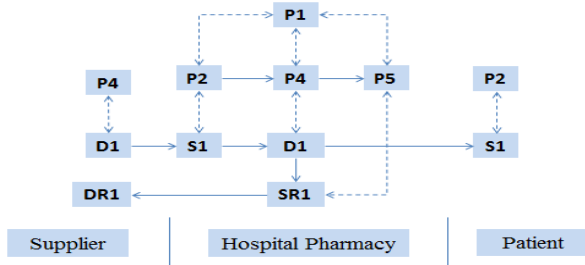


Fig. 1. The tactical level model

C. Operational level

1) Source Stocked Product :

As shown in Fig. 4, S1 is composed of five process elements, Schedule Product Deliveries (S1.1), Receive Product (S1.2), Verify Product (S1.3), Transfer Product (S1.4) and Authorize Supplier payment (ES1.5). S1.1 allows scheduling the drugs delivery; it receives in input the procurement plan from the process Establish Sourcing Plan (P2.4), and the management data of the upstream logistics from the process Manage Incoming Product (ES.6). As output, it gives the drug list to order to the process Identify, Prioritize and Aggregate Product Resources (P2.2). S1.2 allows the drugs reception; it receives in input receipt order from the process Identify, Prioritize and Aggregate Production Requirements (P3.1), and the return plan from the process Establish and Communicate Return Plans (P5.4). As output, it gives the receipt verification data to the process S1.3. S1.3 allows the drugs verification; it receives in input the receipt verification data from S1.2. S1.4 allows drugs delivery; it receives in input the drugs location data in stock from the process Manage Product Inventory ES.4. As output it gives the actual data of drugs stock to the process Manage Finished Goods Inventories (ED.4). S1.5 allows the supplier payment; it receives in input the payment conditions data from the process Manage Supplier Agreements (ES.10).

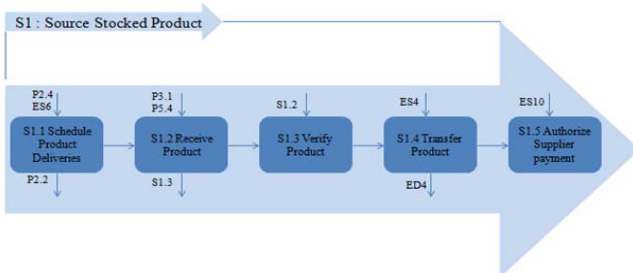


Fig. 2. The Source Stocked Product process model

2) Deliver Stocked Product :

In our case, the distribution of drugs within the hospital is done through five steps, so we choose five process elements which fit most to the hospital management, and which are Process Inquiry and Quote (D1.1), Reserve Inventory and Determine Delivery Date (D1.3), Receive Product from Source or Make (D1.8), Ship Product (D1.12), Receive and Verify Product by Customer (D1.13). D1.1 is the process of answering the information request. D1.3 allows the booking stock and determines the delivery date; it receives in input the data stock availability from the process Manage Deliver Business Rules (ED.1). As output, it gives the delivery date and the stock information to the process Manage Deliver Information (ED.3). D1.8 allows the drugs purchased receipt. As output, it gives the actual data of drugs stock to the processes ES.4 and ED.4. D1.12 allows the drugs delivery. D1.13 allows the receipt and verification of drugs by the patient. The D1 modeling at this level is shown in Fig. 5.

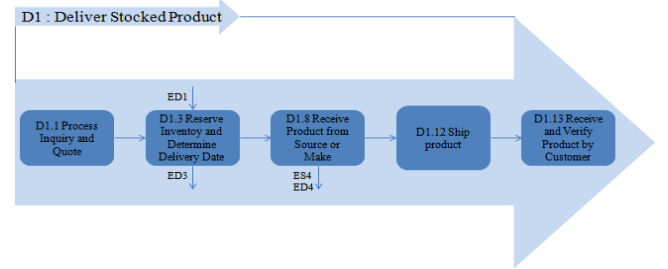


Fig. 4. The Deliver Stocked Product process model

3) Source Return Defective Product :

As shown in Fig. 6, SR1 is composed of five process elements, Identify Defensive Product Condition (SR1.1), Disposition Defective Product (SR1.2), Request Defective Product Return Authorization (SR1.3), Schedule Defective Product Shipment (SR1.4) and Return Defective Product (SR1.5). SR1.1 allows identifying management conditions for defective drugs; it receives in input the return process business rules from the processes Manage Sourcing Business Rules (ES.1) and Manage Business Rules for Return Processes (ER.1). SR1.2 allows the defective drug assignment to the supplier concerned; it receives the return plan from the process Establish and Communicate Return Plans (P5.4). As output, it gives the return authorization to the process D1.8. SR1.3 allows the drug return authorization request; it receives in input the request authorization response from the process Authorize Defective Product Return (DR1.1). As output, it gives the returns rate history to the process Assess and Aggregate Return Requirements (P5.1). SR1.4 allows the scheduling of the defective drugs shipment; it receives in input the stock availability of returns from the process Manage Return Inventory (ER.4). SR1.5 allows the defective drugs return.

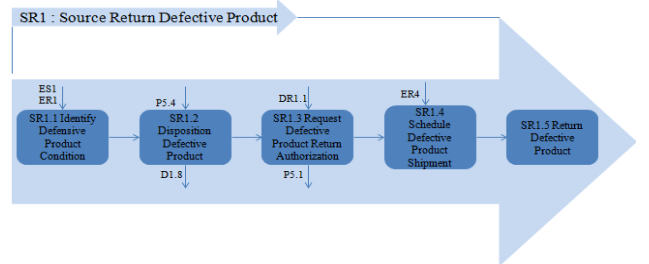


Fig. 3. The Source Return Defective Product process model

D. Real-time level

1) The supply business process model :

The hospital pharmacy pool consists of three lanes: pharmacist, technician and stock manager. The process begins with the start event 'Annual_drug_needs' and ends with the end event 'drugs stored'. After receiving the Annual needs, pharmacists proceeded to the needs study, he prepares the necessary documents to validate this annual needs. After opening the bid, if they were successful, he sends the delivery request. After receiving drugs, the technician updates the data, and the stock manager stored the received drugs. Fig. 7 shows the supply business process model.

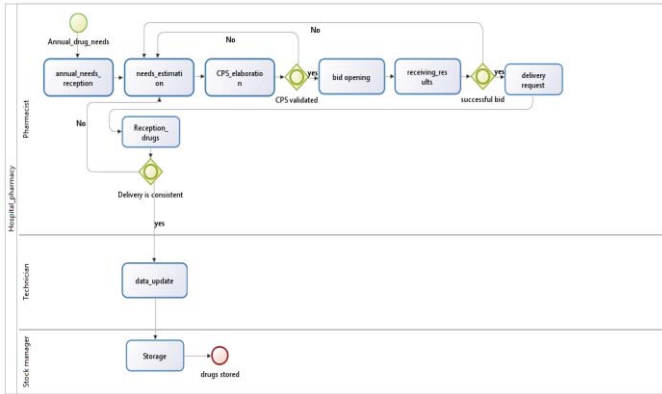


Fig. 7. The supply business process model

2) The distribution business process model :

The hospital pool consists of two lanes: pharmacist and patient. The process begins with the start event 'order receipt' and ends with the end event 'drugs delivered'. After receiving the order receipt, the pharmacist proceeded to the pharmaceutical analysis, allowing him to verify the drugs before updating the data and preparing the drugs to be delivered. Then, the patient receives the drug. Fig. 8 shows the distribution business process model.

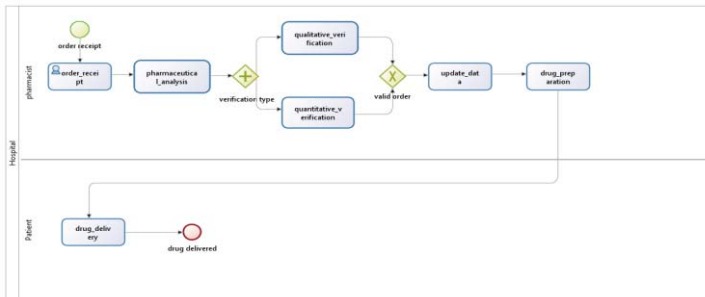


Fig. 8. The distribution business process model

3) The distribution business process model :

The hospital pharmacy pool consists of one lane: pharmacist. The process begins with the start event 'returned drugs' and ends with one of the three end events : 'destroyed drugs', 'sent drugs' and 'returned drugs'. After receiving the returned drugs list, the pharmacist proceeded to the drugs collection, and the elaboration of the necessary documents. Depending on the return type, either the pharmacist proceeds to 'drugs destruction' or 'donations to other hospitals' or 'supplier

return request'. The return business process model is shown by Fig. 9.

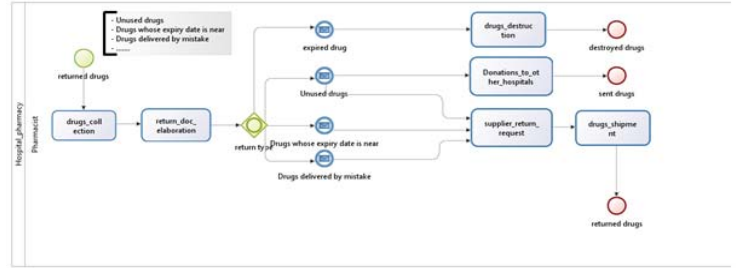


Fig. 9. The return business process model

V. THE PERFORMANCE ANALYSIS

In this part, we propose the Key Performance Indicators (KPI) that are suitable for measuring the supply chain performance. The SCOR model first level metrics characterize performance from customer-facing and internal-facing perspectives. Specific tasks to be completed at the first level are: set business requirements and define basis of competition, evaluate the performance of current operation vis-a-vis required performance, set the SCOR model metrics and targets, and define the gap, set business priorities and state what needs to change [26]. In this work, we focus on the supply chain reliability, for the different levels of the supply chain SCOR provides 18 indicators related to reliability. Therefore, among these indicators proposed by SCOR, we choose those that are most appropriate to the drug management in the hospital, namely: RL.1.1 Perfect order fulfillment, RL.2.1 percentage of orders delivered in full, RL.2.2 Delivery performance to customer commit date [27]. On the other hand, the literature gives a huge number of indicators to evaluate the real-time level of the hospital drug supply chain. Based on these researches [28] [29], we propose the following indicators: Rate out of stock (Allows assessing the level of stocks and quality drug supply, Number of days of stock coverage (Allows assessing the level of stocks based on the consumption), Number of analyzed and validated prescription. Therefore, the set of indicators proposed for assessing the drug supply chain reliability within the hospital is: RL.1.1 Perfect order fulfillment, RL.2.1 percentage of orders delivered in full, RL.2.2 Delivery performance to customer commit date, Rate out of stock, Number of days of stock coverage, Number of analyzed and validated prescription.

VI. CONCLUSION

The work presented in this paper is the result of a literature study, and an analysis and modeling of the hospital drug supply chain that provides a comprehensive view of the various material and information flows in this supply chain. This work results are the mapping of the hospital drug supply chain, in the 3rd level of the SCOR model; the modeling of the main execution processes in BPMN 2.0 through Bonita BPM 7, and a proposition of a set of indicators to evaluate the reliability of a drug supply chain within the hospital. This work will allow us to implement and study the behavior of business processes within the hospital and to develop a dashboard for managing medication in hospital.

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